

Lecture 4: Subspace and Linear Independence

Proving Rank is Well-Defined Through Geometry

Motivation: Is Rank Well-Defined?

Recall: Rank from Cross-Filling (Lecture 3)

In Lecture 3, we defined:

Definition (Rank — Algorithmic Definition)

$\text{rank}(A)$ = number of rank-one pieces from cross-filling

Cross-filling decomposes:

$$A = R_1 + R_2 + \cdots + R_r = UV$$

where each R_i is rank-one, and r is the rank.

The Problem: Different Pivot Choices

Cross-filling allows **different pivot choices** at each step.

$$A = \begin{pmatrix} 2 & 4 & 6 \\ 1 & 2 & 3 \\ 3 & 6 & 9 \end{pmatrix}$$

We could pick pivot at:

- Position (1, 1) with value 2
- Position (2, 2) with value 2
- Position (3, 3) with value 9

Question: Do all choices give the **same number** of pieces?

Today's Goal

Prove that rank is well-defined by discovering:

$$\text{rank}(A) = \text{dimension of column space}$$

Strategy:

1. Understand **two languages** for describing sets
2. Define **subspaces** and **column space**
3. Introduce **linear independence**, **basis**, **dimension**
4. Prove cross-filling produces a **basis** for column space
5. Prove all bases have **same size** (using trace)
6. Conclude: $\text{rank} = \text{dimension}$ (well-defined!)

We elevate rank from **algorithm** to **geometry**.

Two Languages for Describing Sets

Daily Life: Finding Your Classroom

Suppose you want to tell someone **where your classroom is**.

Method A (Descriptive): List properties it satisfies

- "Find the room that has exactly 30 desks"
- "Has windows facing south"
- "Is on the 2nd floor"

Method B (Constructive): Give directions to generate it

- "Go to 2nd floor"
- "Walk to end of hall"
- "Turn left, last door"

Two Languages: Comparison

	Descriptive	Constructive
How	List properties members must satisfy	Provide recipe to generate members
Strength	Easy to verify membership	Easy to generate new members
Weakness	Hard to produce members	Hard to verify membership
Example	"Red hat wearers"	"2nd floor, last door left"

Key: Neither is better — they're **complementary**! We need both.

Example: Line Through Origin in $\mathbb{R}^2$

Consider the line $y = 2x$:

Descriptive language (equation):

$$L = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} : y = 2x \right\}$$

- To **verify** $\begin{pmatrix} 3 \\ 6 \end{pmatrix} \in L$: Check $6 = 2(3)$ ✓
- To **generate** members: Hard! Need to solve equation

Constructive language (parametric):

$$L = \left\{ t \begin{pmatrix} 1 \\ 2 \end{pmatrix} : t \in \mathbb{R} \right\}$$

- To **generate** members: Pick any t ! E.g., $t = 3$ gives $\begin{pmatrix} 3 \\ 6 \end{pmatrix}$
- To **verify** $\begin{pmatrix} 3 \\ 6 \end{pmatrix} \in L$: Hard! Find t such that $t \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 3 \\ 6 \end{pmatrix}$

Example: Plane Through Origin in $\mathbb{R}^3$

Descriptive:

$$P = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} : x + y + z = 0 \right\}$$

Easy to verify: Check if sum = 0

Constructive:

$$P = \left\{ s \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} + t \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} : s, t \in \mathbb{R} \right\}$$

Easy to generate: Pick any s, t

Same plane, two languages!

Connection to Matrices

For a matrix A :

Null space (descriptive):

$$\{x : Ax = 0\}$$

Described by an **equation** that members must satisfy.

Column space (constructive):

$$\{Ax : x \in \mathbb{R}^n\}$$

Described by a **recipe** to generate members.

Solving Equations Overcomes Language Weaknesses

Constructive language weakness: Hard to **verify** if $b \in \text{Col}(A)$

Given b , is it a linear combination of columns of A ? Hard to check directly!

How to compensate? Switch to the **descriptive/equation perspective**:

$$\text{Solve } Ax = b$$

- If solution exists $\rightarrow b \in \text{Col}(A)$ ✓
- If no solution $\rightarrow b \notin \text{Col}(A)$ ✗

Key: **Solving equations** is the **tool** to overcome constructive language's verification weakness!

Not "understanding language helps solve" — it's "solving helps overcome language limitation".

Subspaces: Flat Spaces Through Origin

Definition: Subspace

Definition (Subspace)

A **subspace** W of $\mathbb{R}^n$ is a non-empty subset that is **closed under linear combinations**:

For any $\mathbf{u}, \mathbf{v} \in W$ and any scalars $\alpha, \beta \in \mathbb{R}$:

$$\alpha \mathbf{u} + \beta \mathbf{v} \in W$$

Equivalently, check these three conditions:

1. $\mathbf{0} \in W$ (contains zero vector)
2. If $\mathbf{u}, \mathbf{v} \in W$, then $\mathbf{u} + \mathbf{v} \in W$ (closed under addition)
3. If $\mathbf{u} \in W$ and $\alpha \in \mathbb{R}$, then $\alpha \mathbf{u} \in W$ (closed under scaling)

Geometric Intuition: Flat Spaces

A subspace is a **flat space** that:

- Passes through the **origin** (must contain **0**)
- Is **closed** under vector operations

Examples of subspaces:

- Lines through origin in $\mathbb{R}^2$ or $\mathbb{R}^3$
- Planes through origin in $\mathbb{R}^3$
- All of $\mathbb{R}^n$ itself
- Just $\{\mathbf{0}\}$ (the zero subspace)

Non-examples:

- Line $y = 2x + 1$ (doesn't pass through origin)
- Unit circle (not closed under scaling: $2 \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ not on circle)

Example: Verifying a Subspace

Question: Is $W = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} : x + 2y - z = 0 \right\}$ a subspace of $\mathbb{R}^3$?

Step 1: Check $\mathbf{0} \in W$

$$\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} : 0 + 2(0) - 0 = 0 \quad \checkmark$$

Step 2: Check closure under linear combinations

Take any $\mathbf{u}, \mathbf{v} \in W$. Then:

- $u_1 + 2u_2 - u_3 = 0$ (since $\mathbf{u} \in W$)
- $v_1 + 2v_2 - v_3 = 0$ (since $\mathbf{v} \in W$)

Example: Verifying a Subspace (continued)

For any scalars α, β , check $\alpha \mathbf{u} + \beta \mathbf{v} \in W$:

$$\alpha \mathbf{u} + \beta \mathbf{v} = \begin{pmatrix} \alpha u_1 + \beta v_1 \\ \alpha u_2 + \beta v_2 \\ \alpha u_3 + \beta v_3 \end{pmatrix}$$

Verify the equation:

$$\begin{aligned} & (\alpha u_1 + \beta v_1) + 2(\alpha u_2 + \beta v_2) - (\alpha u_3 + \beta v_3) \\ &= \alpha(u_1 + 2u_2 - u_3) + \beta(v_1 + 2v_2 - v_3) \\ &= \alpha(0) + \beta(0) = 0 \quad \checkmark \end{aligned}$$

Conclusion: W is a subspace! (It's a plane through origin)

Non-Example: Line Not Through Origin

Question: Is $U = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} : x + y = 1 \right\}$ a subspace?

Quick check: Is $\mathbf{0} \in U$?

$$\begin{pmatrix} 0 \\ 0 \end{pmatrix} : 0 + 0 = 0 \neq 1 \quad \times$$

Conclusion: NOT a subspace!

This is a line not through the origin.

Column Space: The Constructive Subspace

Definition: Column Space

Definition (Column Space)

For an $m \times n$ matrix $A = \begin{pmatrix} | & & | \\ \mathbf{a}_1 & \cdots & \mathbf{a}_n \\ | & & | \end{pmatrix}$:

$$\text{Col}(A) = \{Ax : x \in \mathbb{R}^n\}$$

Equivalently (column combination view):

$$\text{Col}(A) = \{x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n : x_1, \dots, x_n \in \mathbb{R}\}$$

This is the set of **all linear combinations of columns** of A .

Language: Column space is **constructive** — defined by a **recipe**!

Pick coefficients $x_1, \dots, x_n$, form the combination.

Column Space is a Subspace

Proposition

$\text{Col}(A)$ is a subspace of $\mathbb{R}^m$.

Proof:

(1) Contains $\mathbf{0}$: Take $x = \mathbf{0}$, then $A\mathbf{0} = \mathbf{0} \in \text{Col}(A)$ ✓

(2) Closed under linear combinations:

- Take $\mathbf{b}_1, \mathbf{b}_2 \in \text{Col}(A)$
- Then $\mathbf{b}_1 = A\mathbf{x}_1$ and $\mathbf{b}_2 = A\mathbf{x}_2$ for some x_1, x_2
- For any α, β :

$$\alpha\mathbf{b}_1 + \beta\mathbf{b}_2 = \alpha A\mathbf{x}_1 + \beta A\mathbf{x}_2 = A(\alpha\mathbf{x}_1 + \beta\mathbf{x}_2) \in \text{Col}(A)$$

✓ Therefore $\text{Col}(A)$ is a subspace!

Example: Column Space of Rank-One Matrix

$$A = \begin{pmatrix} 2 & 4 & 6 \\ 1 & 2 & 3 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \end{pmatrix}$$

All columns are multiples of $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$:

$$\text{Col}(A) = \left\{ t \begin{pmatrix} 2 \\ 1 \end{pmatrix} : t \in \mathbb{R} \right\}$$

This is a **line through origin** in $\mathbb{R}^2$!

Visual: All points on line $y = \frac{1}{2}x$.

Connection to Solvability (Lecture 3 Recall)

Recall from Lecture 3:

$$Ax = b \text{ has a solution} \iff b \in \text{Col}(A)$$

Why?

Because Ax is a linear combination of columns of A , and $\text{Col}(A)$ is **exactly** the set of all such combinations!

Column space captures which vectors can be produced by A .

Linear Independence and Span

Motivation: Redundant Recipe Vectors

Shinchan's coffee shop has recipe vectors (columns = ingredients):

			
	2	1	3
	1	2	3
	0	1	1

Question: Is  redundant?

Notice:  =  +  (ingredient-wise)

$$\begin{pmatrix} 3 \\ 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$$

Yes! We can remove  and still make everything.

This is **linear dependence**.

The Real Question: Minimum Generators

Motivation for Linear Independence:

Problem: Given recipe vectors $v_1, v_2, \dots, v_k$:

- Can we write one using the others? (redundancy)
- Can we **remove** some and still generate everything?
- What's the **minimum set** needed?

Goal: Find vectors with **no redundancy** — each contributes something **new**.

This concept is called **linear independence**.

Definition: Linear Independence

Definition (Linearly Independent)

Vectors $\mathbf{v}_1, \dots, \mathbf{v}_k$ are **linearly independent** if:

$$c_1\mathbf{v}_1 + \dots + c_k\mathbf{v}_k = \mathbf{0} \quad \Rightarrow \quad c_1 = \dots = c_k = 0$$

Intuition: The only way to combine them to get zero is the trivial way (all coefficients zero).

Contrapositive interpretation: No vector can be written as a combination of the others.

Definition (Linearly Dependent)

If there exists a **non-trivial** combination giving zero:

$$c_1\mathbf{v}_1 + \dots + c_k\mathbf{v}_k = \mathbf{0} \quad \text{with some } c_i \neq 0$$

Why This Definition Captures Redundancy

If **dependent**, say $c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k = \mathbf{0}$ with $c_1 \neq 0$:

$$c_1\mathbf{v}_1 = -c_2\mathbf{v}_2 - \cdots - c_k\mathbf{v}_k$$

$$\mathbf{v}_1 = -\frac{c_2}{c_1}\mathbf{v}_2 - \cdots - \frac{c_k}{c_1}\mathbf{v}_k$$

$\mathbf{v}_1$ can be written using the others — **redundant**!

If **independent**, no such representation exists — **no redundancy**.

Example: Testing Independence

Are $\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ linearly independent?

Test: Check $c_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

$$\begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Only solution: $c_1 = 0, c_2 = 0$ ✓

Conclusion: Linearly independent!

Example: Detecting Dependence

Are $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$, $\begin{pmatrix} 4 \\ 2 \end{pmatrix}$, $\begin{pmatrix} 6 \\ 3 \end{pmatrix}$ linearly independent?

Observe: Column 2 = 2·Column 1, Column 3 = 3·Column 1

Try $c_1 = 3$, $c_2 = 0$, $c_3 = -1$:

$$3 \begin{pmatrix} 2 \\ 1 \end{pmatrix} + 0 \begin{pmatrix} 4 \\ 2 \end{pmatrix} - 1 \begin{pmatrix} 6 \\ 3 \end{pmatrix} = \begin{pmatrix} 6 \\ 3 \end{pmatrix} - \begin{pmatrix} 6 \\ 3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Non-trivial solution exists!

Conclusion: Linearly **dependent**!

Definition: Span

Definition (Span)

The **span** of vectors $\mathbf{v}_1, \dots, \mathbf{v}_k$ is the set of all their linear combinations:

$$\text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\} = \{c_1\mathbf{v}_1 + \dots + c_k\mathbf{v}_k : c_1, \dots, c_k \in \mathbb{R}\}$$

We say $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ **spans** W if $\text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\} = W$.

Connection to column space:

$$\text{Col}(A) = \text{span}\{\text{columns of } A\}$$

They're the same thing!

Basis: Minimal Spanning Sets

Definition: Basis

Definition (Basis)

A **basis** for a subspace W is a list of vectors that:

1. **Spans** W (every vector in W can be written as their combination)
2. Is **linearly independent** (no redundancy)

Equivalently: A basis is a **minimal spanning set** or a **maximal independent set**.

Key property: A basis gives **unique coordinates**!

If $\{\mathbf{e}_1, \dots, \mathbf{e}_r\}$ is a basis, then **every** $\mathbf{v} \in W$ can be written **uniquely** as:

$$\mathbf{v} = c_1\mathbf{e}_1 + \dots + c_r\mathbf{e}_r$$

Why Basis Gives Unique Coordinates

Existence: Spanning ensures you can write $\mathbf{v}$ as a combination.

Uniqueness: Suppose two representations:

$$\mathbf{v} = c_1\mathbf{e}_1 + \cdots + c_r\mathbf{e}_r = d_1\mathbf{e}_1 + \cdots + d_r\mathbf{e}_r$$

Then:

$$(c_1 - d_1)\mathbf{e}_1 + \cdots + (c_r - d_r)\mathbf{e}_r = \mathbf{0}$$

By linear independence, all $c_i - d_i = 0$, so $c_i = d_i$ for all i . ✓

Basis = efficient coordinate system with no redundancy!

Example: Standard Basis for $\mathbb{R}^3$

$$\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{e}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

Check independence:

$$c_1\mathbf{e}_1 + c_2\mathbf{e}_2 + c_3\mathbf{e}_3 = \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = \mathbf{0}$$

Only if all $c_i = 0$ ✓

Check spanning: Any $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = x\mathbf{e}_1 + y\mathbf{e}_2 + z\mathbf{e}_3$ ✓

Conclusion: $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ is a basis for $\mathbb{R}^3$!

Example: Another Basis for $\mathbb{R}^2$

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

Check independence: From $c_1\mathbf{v}_1 + c_2\mathbf{v}_2 = \mathbf{0}$:

$$c_1 + c_2 = 0, \quad c_1 - c_2 = 0$$

Add: $2c_1 = 0 \Rightarrow c_1 = 0 \Rightarrow c_2 = 0 \checkmark$

Check spanning: For any $\begin{pmatrix} x \\ y \end{pmatrix}$, solve:

$$c_1 + c_2 = x, \quad c_1 - c_2 = y$$

Solution: $c_1 = \frac{x+y}{2}, c_2 = \frac{x-y}{2}$ (always exists!) $\checkmark$

Conclusion: $\{\mathbf{v}_1, \mathbf{v}_2\}$ is another basis for $\mathbb{R}^2$!

Left Cancellation Property

Matrix Form of Independence

Collect vectors $v_1, \dots, v_k$ as columns of matrix U :

$$U = \begin{pmatrix} | & | & \cdots & | \\ v_1 & v_2 & \cdots & v_k \\ | & | & \cdots & | \end{pmatrix}$$

Linear independence says:

$$U \begin{pmatrix} c_1 \\ \vdots \\ c_k \end{pmatrix} = \mathbf{0} \quad \Rightarrow \quad \begin{pmatrix} c_1 \\ \vdots \\ c_k \end{pmatrix} = \mathbf{0}$$

In other words: $Uc = \mathbf{0} \Rightarrow c = \mathbf{0}$

Left Cancellation Property — Setup

Proposition (Left Cancellation)

If U has linearly independent columns, then:

$$UP = UQ \Rightarrow P = Q$$

Let's write out what each matrix looks like:

$$U = \begin{pmatrix} | & | & \cdots & | \\ \mathbf{u}_1 & \mathbf{u}_2 & \cdots & \mathbf{u}_k \\ | & | & & | \end{pmatrix}$$

$$P = \begin{pmatrix} | & | & \cdots & | \\ \mathbf{p}_1 & \mathbf{p}_2 & \cdots & \mathbf{p}_m \\ | & | & & | \end{pmatrix}, \quad Q = \begin{pmatrix} | & | & \cdots & | \\ \mathbf{q}_1 & \mathbf{q}_2 & \cdots & \mathbf{q}_m \\ | & | & & | \end{pmatrix}$$

U is $n \times k$, P and Q are $k \times m$.

Left Cancellation — What Does UP Mean?

$$UP = \begin{pmatrix} | & | & \cdots & | \\ \mathbf{u}_1 & \mathbf{u}_2 & \cdots & \mathbf{u}_k \\ | & | & \cdots & | \end{pmatrix} \begin{pmatrix} | & | & \cdots & | \\ \mathbf{p}_1 & \mathbf{p}_2 & \cdots & \mathbf{p}_m \\ | & | & \cdots & | \end{pmatrix}$$

Column j of UP is:

$$U\mathbf{p}_j = p_{1j}\mathbf{u}_1 + p_{2j}\mathbf{u}_2 + \cdots + p_{kj}\mathbf{u}_k$$

This is a **linear combination of U 's columns** with coefficients from $\mathbf{p}_j$!

So:

$$UP = \begin{pmatrix} | & | & \cdots & | \\ U\mathbf{p}_1 & U\mathbf{p}_2 & \cdots & U\mathbf{p}_m \\ | & | & \cdots & | \end{pmatrix}$$

Left Cancellation — Column-by-Column

Similarly:

$$UQ = \begin{pmatrix} | & | & \cdots & | \\ U\mathbf{q}_1 & U\mathbf{q}_2 & \cdots & U\mathbf{q}_m \\ | & | & & | \end{pmatrix}$$

If $UP = UQ$, then **column-by-column**:

$$U\mathbf{p}_j = U\mathbf{q}_j \quad \text{for each } j = 1, 2, \dots, m$$

Rearranging:

$$U\mathbf{p}_j - U\mathbf{q}_j = \mathbf{0}$$

$$U(\mathbf{p}_j - \mathbf{q}_j) = \mathbf{0}$$

Left Cancellation — Using Independence

For each column j , we have:

$$U(\mathbf{p}_j - \mathbf{q}_j) = \mathbf{0}$$

Expanding:

$$(\mathbf{p}_j - \mathbf{q}_j) = \begin{pmatrix} p_{1j} - q_{1j} \\ p_{2j} - q_{2j} \\ \vdots \\ p_{kj} - q_{kj} \end{pmatrix}$$

$$U(\mathbf{p}_j - \mathbf{q}_j) = (p_{1j} - q_{1j})\mathbf{u}_1 + (p_{2j} - q_{2j})\mathbf{u}_2 + \cdots + (p_{kj} - q_{kj})\mathbf{u}_k = \mathbf{0}$$

Since $\{\mathbf{u}_1, \dots, \mathbf{u}_k\}$ are **linearly independent**:

$$p_{1j} - q_{1j} = 0, \quad p_{2j} - q_{2j} = 0, \quad \dots, \quad p_{kj} - q_{kj} = 0$$

Left Cancellation — Conclusion

Therefore, for each j :

$$\mathbf{p}_j = \mathbf{q}_j$$

Since this holds for **all columns** $j = 1, 2, \dots, m$:

$$P = \begin{pmatrix} | & | & \cdots & | \\ \mathbf{p}_1 & \mathbf{p}_2 & \cdots & \mathbf{p}_m \\ | & | & \cdots & | \end{pmatrix} = \begin{pmatrix} | & | & \cdots & | \\ \mathbf{q}_1 & \mathbf{q}_2 & \cdots & \mathbf{q}_m \\ | & | & \cdots & | \end{pmatrix} = Q$$

$$\boxed{P = Q}$$

□

Key step: Independence of U 's columns forces each coefficient difference to be zero.

Why "Left" Cancellation?

The matrix U appears on the **left** in $UP = UQ$, and we can "cancel" it.

Contrast: If columns are **dependent**, we **cannot** cancel!

Example:

$$\begin{pmatrix} 1 & 2 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 0 \\ 0.5 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

But $\begin{pmatrix} 1 \\ 0 \end{pmatrix} \neq \begin{pmatrix} 0 \\ 0.5 \end{pmatrix}!$

Cancellation **fails** because columns are dependent.

Consequence: Unique Representation

If $\{v_1, \dots, v_k\}$ are linearly independent and:

$$w = c_1 v_1 + \dots + c_k v_k$$

then the coefficients $c_1, \dots, c_k$ are **unique**.

Proof: If also $w = d_1 v_1 + \dots + d_k v_k$, then in matrix form:

$$U \begin{pmatrix} c_1 \\ \vdots \\ c_k \end{pmatrix} = U \begin{pmatrix} d_1 \\ \vdots \\ d_k \end{pmatrix}$$

By left cancellation: $c_i = d_i$ for all i . $\square$

Cross-Filling Produces a Basis

Recall: Cross-Filling (Lecture 3)

Cross-filling decomposes:

$$A = R_1 + R_2 + \cdots + R_r = UV$$

where:

- Each $R_i = \mathbf{u}_i \mathbf{v}_i^T$ is rank-one
- $U = \begin{pmatrix} | & & | \\ \mathbf{u}_1 & \cdots & \mathbf{u}_r \\ | & & | \end{pmatrix}$ collects the column vectors
- $V = \begin{pmatrix} - & \mathbf{v}_1^T & - \\ & \vdots & \\ - & \mathbf{v}_r^T & - \end{pmatrix}$ collects the row vectors

Key question: What is special about U 's columns?

Property 1: U 's Columns Are Independent

Proposition

Let $A = UV$ from cross-filling. Then the columns of U are **linearly independent**.

Proof idea: Use the **pivot structure**!

Each $\mathbf{u}_i$ came from a rank-one piece with pivot in some row p_i .

After peeling $R_1, \dots, R_{i-1}$, rows $p_1, \dots, p_{i-1}$ of the remainder are **zero**.

This creates a **triangular pattern** that forces independence.

Concrete Example: Rank-3 with Off-Diagonal Pivots

Goal: Show cross-filling with **pivots NOT on diagonal**, using **3 colors**.

Let's do cross-filling on a rank-3 matrix:

$$A = \begin{pmatrix} 0 & 6 & 3 \\ 0 & 2 & 4 \\ 5 & 1 & 7 \end{pmatrix}$$

Pivots will be at:

- **Pivot 1:** Row 1, Col 2 — **Red**
- **Pivot 2:** Row 2, Col 3 — **Blue**
- **Pivot 3:** Row 3, Col 1 — **Green**

Notice: NOT (1,1), (2,2), (3,3) — completely **off diagonal**!

Step 1: First Pivot (Row 1, Col 2) — Red

Pick pivot at position (row 1, column 2), value 6:

$$\text{Row 1} \begin{pmatrix} 0 & \boxed{6} & 3 \\ 0 & 2 & 4 \\ 5 & 1 & 7 \end{pmatrix} \quad \text{Pivot 1}$$

Extract R_1 :

$$R_1 = \underbrace{\begin{pmatrix} 6 \\ 2 \\ 1 \end{pmatrix}}_{u_1 = \text{column 2}} \cdot \underbrace{\begin{pmatrix} 0 & 1 & \frac{1}{2} \end{pmatrix}}_{v_1^T = \text{row 1/6}} = \begin{pmatrix} 0 & 6 & 3 \\ 0 & 2 & 1 \\ 0 & 1 & \frac{1}{2} \end{pmatrix}$$

Step 2: After Peeling R_1 — Row 1 Eliminated

Remainder:

$$A_2 = A - R_1 = \begin{pmatrix} 0 & 6 & 3 \\ 0 & 2 & 4 \\ 5 & 1 & 7 \end{pmatrix} - \begin{pmatrix} 0 & 6 & 3 \\ 0 & 2 & 1 \\ 0 & 1 & \frac{1}{2} \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 3 \\ 5 & 0 & \frac{13}{2} \end{pmatrix}$$

Row 1 is ZERO! 

$$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 3 \\ 5 & 0 & \frac{13}{2} \end{pmatrix}$$

Pivot 2

Key: After peeling R_1 , row 1 (the pivot row) became zero!

New pivot: (row 2, col 3), value 3

Step 3: Extract R_2 — Notice the Zeros!

From remainder $A_2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 3 \\ 5 & 0 & \frac{13}{2} \end{pmatrix}$, pivot at (row 2, col 3):

Extract R_2 :

$$R_2 = \underbrace{\begin{pmatrix} 0 \\ 3 \\ \frac{13}{2} \end{pmatrix}}_{\mathbf{u}_2 = \text{column 3}} \cdot \underbrace{\begin{pmatrix} 0 & 0 & 1 \end{pmatrix}}_{\mathbf{v}_2^T = \text{row 2/3}}$$

Observe $\mathbf{u}_2$:

$$\mathbf{u}_2 = \begin{pmatrix} 0 \\ 3 \\ \frac{13}{2} \end{pmatrix} \leftarrow \text{First entry is 0!}$$

Why? Because row 1 of A_2 was already zero (eliminated by R_1).

So column 3 of A_2 must have 0 in row 1!

Step 4: After Peeling R_2 — Row 2 Eliminated

Remainder after R_2 :

$$A_3 = A_2 - R_2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 3 \\ 5 & 0 & \frac{13}{2} \end{pmatrix} - \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 3 \\ 0 & 0 & \frac{13}{2} \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 5 & 0 & 0 \end{pmatrix}$$

Row 1 zero
Row 2 zero

Pivot 3

$$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 5 & 0 & 0 \end{pmatrix}$$

Now: Row 1 AND Row 2 are zero!

Final pivot: (row 3, col 1), value 5

Step 5: Extract R_3 — All Pivots Found!

From remainder $A_3 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 5 & 0 & 0 \end{pmatrix}$, pivot at (row 3, col 1):

Extract R_3 :

$$R_3 = \underbrace{\begin{pmatrix} 0 \\ 0 \\ 5 \end{pmatrix}}_{\mathbf{u}_3 = \text{column 1}} \cdot \underbrace{\begin{pmatrix} 1 & 0 & 0 \end{pmatrix}}_{\mathbf{v}_3^T = \text{row 3/5}}$$

Observe $\mathbf{u}_3$:

$$\mathbf{u}_3 = \begin{pmatrix} 0 \\ 0 \\ 5 \end{pmatrix} \leftarrow \text{First TWO entries are } 0, 0!$$

Why? Because row 1 and row 2 of A_3 were both zero.

Column 1 of A_3 has 0 in row 1, 0 in row 2!

The Perfect Triangular Structure — 3 Colors!

Collected columns from cross-filling:

$$U = \begin{pmatrix} \mathbf{u}_1 & \mathbf{u}_2 & \mathbf{u}_3 \end{pmatrix} = \begin{pmatrix} 6 & 0 & 0 \\ 2 & 3 & 0 \\ 1 & \frac{13}{2} & 5 \end{pmatrix}$$

The diagram shows the matrix U with its columns highlighted in different colors: the first column is red, the second is blue, and the third is green. Red arrows point from the zero entries in the first row to the text "Zero (row 1 eliminated)". A blue arrow points from the zero entry in the second row to the text "Zero (row 2 eliminated)".

Row 1 $\begin{pmatrix} 6 & 0 & 0 \end{pmatrix}$ Zero (row 1 eliminated)

Row 2 $\begin{pmatrix} 2 & 3 & 0 \end{pmatrix}$ Zero (row 2 eliminated)

Row 3 $\begin{pmatrix} 1 & \frac{13}{2} & 5 \end{pmatrix}$

Perfect triangular pattern:

- $\mathbf{u}_2$ has 0 in row 1 (previous pivot row)
- $\mathbf{u}_3$ has 0 in row 1, 0 in row 2 (all previous pivot rows)

Visual Summary: Why Independence? — Backward Substitution

$$U = \begin{pmatrix} 6 & 0 & 0 \\ 2 & 3 & 0 \\ 1 & \frac{13}{2} & 5 \end{pmatrix}$$

Check $c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + c_3\mathbf{u}_3 = \mathbf{0}$:

- **Row 1:** $6c_1 + 0c_2 + 0c_3 = 0 \Rightarrow \boxed{c_1 = 0}$
- **Row 2:** $2c_1 + 3c_2 + 0c_3 = 0 + 3c_2 + 0 = 0 \Rightarrow \boxed{c_2 = 0}$
- **Row 3:** $1c_1 + \frac{13}{2}c_2 + 5c_3 = 0 + 0 + 5c_3 = 0 \Rightarrow \boxed{c_3 = 0}$

Triangular pattern (colored zeros!) allows **backward substitution**:
solve row 1 $\rightarrow$ row 2 $\rightarrow$ row 3!

General Proof: Pivot Row Method

Let $\mathbf{u}_1, \dots, \mathbf{u}_r$ be columns from cross-filling, where $\mathbf{u}_i$ has pivot in row p_i .

Key observation: After peeling $R_1, \dots, R_{i-1}$, rows $p_1, \dots, p_{i-1}$ of remainder are zero.

Therefore:

$$\mathbf{u}_i[\text{row } p_j] = 0 \quad \text{for all } j < i$$

Check independence: Suppose $c_1\mathbf{u}_1 + \dots + c_r\mathbf{u}_r = \mathbf{0}$.

Solve backwards:

1. Look at pivot row p_r : Only $\mathbf{u}_r$ has nonzero entry there, so $c_r = 0$
2. Look at pivot row p_{r-1} : Only $\mathbf{u}_{r-1}$ and $\mathbf{u}_r$ could contribute, but $c_r = 0$, so $c_{r-1} = 0$
3. Continue: $c_{r-2} = 0, \dots, c_1 = 0$

All coefficients zero! ✓

Property 2: $\text{Col}(U) = \text{Col}(A)$

Proposition

If $A = UV$ from cross-filling, then:

$$\text{Col}(U) = \text{Col}(A)$$

Proof (Two inclusions):

Part 1: $\text{Col}(A) \subseteq \text{Col}(U)$

Since $A = UV$:

$$Ax = UVx = U(Vx)$$

Let $y = Vx$, then $Ax = Uy \in \text{Col}(U)$ ✓

Property 2 (continued): The Deeper Direction

Part 2: $\text{Col}(U) \subseteq \text{Col}(A)$

Must show: Every column of U is in $\text{Col}(A)$.

Key idea: Cross-filling **peels columns from A !**

Each $\mathbf{u}_i$ comes from a column of some remainder matrix A_k .

Remainders are built from A 's columns, so $\mathbf{u}_i \in \text{Col}(A)$.

Detailed Argument: Induction on Remainders

Cross-filling process:

- Start: $A_1 = A$
- Peel $R_1 = \mathbf{u}_1 \mathbf{v}_1^T$ where $\mathbf{u}_1$ is (scaled) column of A_1
- Remainder: $A_2 = A_1 - R_1$
- Peel $R_2 = \mathbf{u}_2 \mathbf{v}_2^T$ where $\mathbf{u}_2$ is (scaled) column of A_2
- Continue...

Claim: Every column of A_k is in $\text{Col}(A)$.

Proof by induction:

- **Base:** $A_1 = A$ ✓
- **Step:** $A_{k+1} = A_k - \mathbf{u}_k \mathbf{v}_k^T$
Each column: $(A_k)_j - v_{kj} \mathbf{u}_k = \text{combination of } A_k \text{'s columns}$
By induction, both $(A_k)_j$ and $\mathbf{u}_k$ are in $\text{Col}(A)$
So their combination is too! ✓

Conclusion: Cross-Filling Extracts a Basis

Theorem (Main Result)

If $A = UV$ from cross-filling, then:

The columns of U form a basis for $\text{Col}(A)$!

Proof:

1. ✓ **Linearly independent** (pivot structure)
2. ✓ **Spans $\text{Col}(A)$** ($\text{Col}(U) = \text{Col}(A)$)

Therefore $\{\mathbf{u}_1, \dots, \mathbf{u}_r\}$ is a basis! $\square$

Trace and Dimension

Well-Definedness

The Dimension Question

Definition (Dimension — Tentative)

If W has a basis with r vectors, we define:

$$\dim(W) = r$$

Problem: What if W has two different bases with **different sizes**?

- Basis 1: $\{\mathbf{u}_1, \dots, \mathbf{u}_m\}$ (m vectors)
- Basis 2: $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ (n vectors)

Is $m = n$?

Goal: Prove all bases have the same size!

The Tool: Trace

Definition (Trace)

The **trace** of a square matrix A is the sum of diagonal entries:

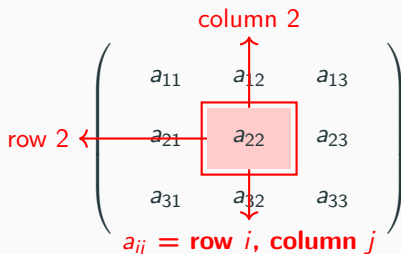
$$\text{tr}(A) = a_{11} + a_{22} + \cdots + a_{nn}$$

Example:

$$\text{tr} \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} = 1 + 5 + 9 = 15$$

Key property: $\text{tr}(AB) = \text{tr}(BA)$ even if A, B not square!

Subscript Notation: What Does a_{ij} Mean?



A 3x3 matrix is shown with elements a_{11}, a_{12}, a_{13} in the first row, a_{21}, a_{22}, a_{23} in the second row, and a_{31}, a_{32}, a_{33} in the third row. The element a_{22} is highlighted with a red square. A red arrow points from the text "column 2" to a_{22} , and another red arrow points from the text "row 2" to a_{22} . Below the matrix, the text $a_{ij} = \text{row } i, \text{ column } j$ is written in red.

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

$a_{ij} = \text{row } i, \text{ column } j$

For a_{22} : first index = row number (2), second index = column number (2).

Visual Proof: $\text{tr}(AB) = \text{tr}(BA)$ — Setup

Consider A is 2×3 and B is 3×2 :

$$\begin{matrix} & A_{2 \times 3} & & B_{3 \times 2} \\ \left(\begin{array}{|c|c|c|} \hline a_{11} & a_{12} & a_{13} \\ \hline a_{21} & a_{22} & a_{23} \\ \hline \end{array} \right) & & \left(\begin{array}{|c|c|} \hline b_{11} & b_{12} \\ \hline b_{21} & b_{22} \\ \hline b_{31} & b_{32} \\ \hline \end{array} \right) \end{matrix}$$

First, compute AB (will be 2×2):

Diagonal entries:

$$(AB)_{11} = a_{11}b_{11} + a_{12}b_{21} + a_{13}b_{31}$$

$$(AB)_{22} = a_{21}b_{12} + a_{22}b_{22} + a_{23}b_{32}$$

Visual Proof: $\text{tr}(AB)$

$$\text{tr}(AB) = (AB)_{11} + (AB)_{22}$$

$$\begin{aligned} &= (a_{11}b_{11} + a_{12}b_{21} + a_{13}b_{31}) \\ &\quad + (a_{21}b_{12} + a_{22}b_{22} + a_{23}b_{32}) \end{aligned}$$

This is the sum of all **color-matched pairs**.

Visual Proof: Now Compute BA

BA is 3×3 :

$$\begin{matrix} & B & & A \\ \left(\begin{array}{cc|c} b_{11} & b_{12} & \\ b_{21} & b_{22} & \\ b_{31} & b_{32} & \end{array} \right) & \left(\begin{array}{ccc} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{array} \right) \end{matrix}$$

Diagonal entries of BA :

$$(BA)_{11} = b_{11}a_{11} + b_{12}a_{21}$$

$$(BA)_{22} = b_{21}a_{12} + b_{22}a_{22}$$

$$(BA)_{33} = b_{31}a_{13} + b_{32}a_{23}$$

Visual Proof: $\text{tr}(BA)$

$$\text{tr}(BA) = (BA)_{11} + (BA)_{22} + (BA)_{33}$$

$$\begin{aligned} &= (b_{11}a_{11} + b_{12}a_{21}) \\ &\quad + (b_{21}a_{12} + b_{22}a_{22}) \\ &\quad + (b_{31}a_{13} + b_{32}a_{23}) \end{aligned}$$

By commutativity of multiplication ($a_{ij}b_{kl} = b_{kl}a_{ij}$):

$$= (a_{11}b_{11} + a_{21}b_{12}) + (a_{12}b_{21} + a_{22}b_{22}) + (a_{13}b_{31} + a_{23}b_{32})$$

This is **exactly the same** sum as $\text{tr}(AB)$!

$$\text{tr}(AB) = \text{tr}(BA)$$

Using Trace to Prove Dimension Well-Defined

Theorem

If W has two bases with m and n vectors, then $m = n$.

Proof strategy:

1. Write one basis in terms of the other (change-of-basis matrices)
2. Use left cancellation: these matrices compose to identity
3. Use trace to count sizes

Proof: Change-of-Basis Matrices

Let:

- Basis 1: $U = \begin{pmatrix} | & & | \\ \mathbf{u}_1 & \cdots & \mathbf{u}_m \\ | & & | \end{pmatrix}$
- Basis 2: $V = \begin{pmatrix} | & & | \\ \mathbf{v}_1 & \cdots & \mathbf{v}_n \\ | & & | \end{pmatrix}$

Since Basis 2 **spans** W , each $\mathbf{u}_j$ can be written using $\mathbf{v}_i$'s:

$$U = VP$$

for some $n \times m$ matrix $P = \begin{pmatrix} | & & | \\ \mathbf{p}_1 & \cdots & \mathbf{p}_m \\ | & & | \end{pmatrix}$.

Column $\mathbf{p}_j$ provides **coefficients** to write $\mathbf{u}_j$ as combination of $\mathbf{v}_i$'s!

Proof: The Reverse Direction

Similarly, since Basis 1 **spans** W :

$$V = UQ$$

for some $m \times n$ matrix $Q = \begin{pmatrix} | & & | \\ \mathbf{q}_1 & \cdots & \mathbf{q}_n \\ | & & | \end{pmatrix}$.

Column $\mathbf{q}_i$ provides **coefficients** to write $\mathbf{v}_i$ as combination of $\mathbf{u}_j$'s!

Proof: Composing the Changes

Substitute $V = UQ$ into $U = VP$:

$$U = VP = (UQ)P = U(QP)$$

Since U has **linearly independent** columns, by **left cancellation**:

$$QP = I_m$$

(an $m \times m$ identity matrix)

Similarly, substitute $U = VP$ into $V = UQ$:

$$V = UQ = (VP)Q = V(PQ)$$

By left cancellation:

$$PQ = I_n$$

Proof: Using Trace

We have:

- $QP = I_m$ (size $m \times m$)
- $PQ = I_n$ (size $n \times n$)

Taking traces:

$$\text{tr}(QP) = \text{tr}(I_m) = m$$

$$\text{tr}(PQ) = \text{tr}(I_n) = n$$

But by the trace property:

$$\text{tr}(QP) = \text{tr}(PQ)$$

Therefore: $m = n$ $\square$

Any two bases have the same size!

Dimension is Now Well-Defined

Definition (Dimension)

The **dimension** of a subspace W is the number of vectors in any basis:

$$\dim(W) = \text{number of vectors in any basis for } W$$

This is **independent of which basis** we choose!

Examples:

- $\dim(\mathbb{R}^n) = n$ (standard basis has n vectors)
- $\dim(\text{Col}(A)) = \text{number of vectors in any basis for column space}$
- $\dim(\{\mathbf{0}\}) = 0$ (zero subspace)

Rank is Well-Defined

Connecting Rank to Dimension

Theorem (Rank = Dimension)

For any matrix A :

$$\text{rank}(A) = \dim(\text{Col}(A))$$

Proof:

From §8, cross-filling produces $A = UV$ where:

- U has r columns
- Columns of U form a **basis** for $\text{Col}(A)$
- Therefore $\dim(\text{Col}(A)) = r$

By Lecture 3 definition:

$$\text{rank}(A) = r \text{ (number of rank-one pieces)}$$

Therefore: $\text{rank}(A) = \dim(\text{Col}(A)) \quad \square$

Rank is Well-Defined!

Why? Because:

1. $\text{rank}(A) = \dim(\text{Col}(A))$ (just proved)
2. $\dim(\text{Col}(A))$ is well-defined (all bases have same size)
3. Therefore $\text{rank}(A)$ is well-defined!

Different cross-filling runs (different pivot choices) may give:

- Different U matrices
- Different V matrices
- Different rank-one pieces $R_1, R_2, \dots$

But they all give:

- **Same number r of pieces** (since $r = \dim(\text{Col}(A))$)
- **Same column space** $\text{Col}(A)$

Example: Independence of Pivot Choices

$$A = \begin{pmatrix} 2 & 4 \\ 1 & 2 \end{pmatrix}$$

Pivot choice 1 (position (1,1)):

$$A = \begin{pmatrix} 2 \\ 1 \end{pmatrix} \begin{pmatrix} 1 & 2 \end{pmatrix}$$

One piece $\rightarrow$ rank = 1

Pivot choice 2 (position (2,2)):

$$A = \begin{pmatrix} 1 \\ 0.5 \end{pmatrix} \begin{pmatrix} 2 & 4 \end{pmatrix}$$

One piece $\rightarrow$ rank = 1

Both give rank 1 because:

$$\text{Col}(A) = \left\{ t \begin{pmatrix} 2 \\ 1 \end{pmatrix} : t \in \mathbb{R} \right\}$$

Summary

What We Achieved Today

From Algorithm to Geometry:

Lecture 3: Rank was **algorithmic** (count pieces from cross-filling)

- Problem: Different pivot choices might give different counts

Lecture 4: Rank is **geometric**:

$$\text{rank}(A) = \dim(\text{Col}(A))$$

- Solution: Dimension is well-defined (all bases same size)
- Therefore rank is well-defined!

Key insight: Cross-filling **finds a basis** for $\text{Col}(A)$!

The Conceptual Framework

	Descriptive	Constructive
Definition	List constraints	Give recipe
Example	Null space: $\{x : Ax = 0\}$	Column space: $\{Ax : x \in \mathbb{R}^n\}$
Strength	Verify membership	Generate members

Basis = minimal spanning set = maximal independent set

Dimension = size of any basis (well-defined!)

Rank = dimension of column space = number of basis vectors from cross-filling

Tools We Developed

1. **Two Languages** for sets: Descriptive $\leftrightarrow$ Constructive
2. **Subspaces**: Flat spaces through origin, closed under linear combinations
3. **Linear Independence**: No redundancy, minimum generators
4. **Basis**: Linearly independent + spans = unique coordinates
5. **Left Cancellation**: $UP = UQ \Rightarrow P = Q$ (for independent U)
6. **Cross-filling produces basis**: Pivot structure $\rightarrow$ independence
7. **Trace property**: $\text{tr}(AB) = \text{tr}(BA)$ (visual proof)
8. **Dimension well-defined**: All bases same size (using trace)
9. **Rank = Dimension**: Geometric meaning of rank